

Traditional Flare Detection Methods

Threshold Methods: The canonical approach (used by NOAA SWPC) is to trigger on soft X-ray (SXR) flux exceeding fixed levels. For example, NOAA's GOES algorithm requires the 1–8 Å flux to rise monotonically for 4 min and exceed a 40% increase relative to 3 min prior [13†L763-L770]. This effectively implements C/M/X class thresholds (10^{-6} , 10^{-5} , 10^{-4} W/m²) with a required rise rate. Similarly, RHESSI's flare list used an HXR threshold: it flags a candidate when the 6–12 keV count rate rises $>3\sigma$ above a 60 s background [28†L11-L19] and the front-segment count ratio is $>3\sigma$ [28†L20-L29]. Threshold methods are simple and low-latency (trigger within minutes of onset) but **fail if the flux profile is irregular or high background**. For example, GOES may miss small flares on top of decaying backgrounds [13†L763-L770]. Thresholding is blind to spectral shape (nonthermal content) and has trouble separating closely-timed events (double flares), as discussed in the LYRA/PROBA2 study [69†L67-L75].

Derivative/Rate-of-Rise Methods: Some algorithms trigger on the time derivative of the flux. A rapid dF/dt signals an impulsive phase. For instance, the Solar Ultraviolet Imager (SUVI) triggers flare alerts on steep EUV flux rises. In SXR/HXR, one could set dF/dt thresholds to catch sudden increases. Such methods can respond faster than pure thresholding, but they may generate false alarms from noise spikes or bursty data. (We found no well-known solar-flare papers exclusively using derivative triggers, but they are implicit in algorithms that require “monotonic increase.”)

Bayesian/Statistical Methods: Bayesian change-point methods (e.g. Scargle's Bayesian Blocks) segment time series into piecewise-constant blocks, identifying statistical jumps [31†L173-L182]. In principle this can detect flare onsets as change points in photon counts. Scargle et al. (2013) show Bayesian Blocks can run in a real-time “trigger mode” for event detection [31†L173-L182]. Such methods are optimal for Poisson-count data and can approach the theoretical detection limit, but they assume known noise statistics. In practice, solar X-ray data have complicated backgrounds, so Bayesian Blocks may require tuning. Pitkin et al. (2014) applied a Bayesian odds-ratio detection to stellar white-light flares with an exponential-decay model, greatly reducing false negatives compared to a fixed threshold; a similar approach could, in principle, be applied to solar HXR [18†L41-L49]. Bayesian detectors can be robust to noise but may be computationally heavier and sensitive to model assumptions (flare shape priors).

Change-Point Detection: Related to Bayesian Blocks, general change-point algorithms can flag sudden shifts in mean or variance. These have been used for (e.g.) detecting geomagnetic disturbances, but in solar X-rays they are not widespread in the literature. They share strengths with Bayesian methods (model-agnostic) but can give false triggers on gradual trends if not carefully tuned.

Wavelet Methods: Wavelet transforms decompose the time series by scale. They have been used extensively to analyze quasi-periodic pulsations (QPPs) in flares, and to extract multi-scale features in X-ray time series [76†L1-L9] . A possible detection strategy is to monitor the wavelet power at short timescales: a significant rise in high-frequency wavelet energy could flag an impulsive onset. Wavelet filters (e.g. Mexican-hat) can act as matched filters for impulses. The advantage is multi-resolution sensitivity (detects both slow rise and fast spikes). The drawback is computational cost and the need to choose mother wavelets/scales. As far as we found, no standard operational flare alarm uses wavelets directly; they are more often used in post-facto analysis of flare oscillations.

Matched Filtering / Template Methods: One could cross-correlate the X-ray series with an assumed flare profile (e.g. a Gaussian or two-phase shape). This is analogous to matched filtering in signal processing. It can be optimal if the flare shape is known, but flares vary widely. In practice, we found no solar reference explicitly using matched filters for detection, likely due to shape variability and limited speed.

Signal Processing Techniques: More exotic methods (e.g. Kalman filtering, cumulant analysis) have been rarely applied. In general, any method that highlights sudden non-stationarity can detect flares, but few have been benchmarked. We note that traditional NOAA “event lists” are essentially threshold on a smoothed time series, which is a basic filter. Advanced filters (e.g. adaptive filters) could in principle improve SNR, but literature on this is scant.

Traditional Methods – Summary: Table 1 compares representative approaches. Notably, GOES-based thresholding [13†L763-L770] and RHESSI HXR triggers [28†L11-L19] [28†L20-L29] are proven in operations. Bayesian Blocks [31†L173-L182] offer theoretical advantages in change-point detection. Overall, traditional methods are low-latency and simple, but **have weaknesses:** they assume monotonic or simple flare shapes, ignore multi-wavelength correlations, and often detect flares only minutes after onset (GOES’s 4 min rule imposes delay). They also generate false alarms for noise fluctuations and tend to miss “hot onset” phases when flux changes are gradual [37†L68-L72] .

Table 1: Traditional flare detection methods.

Method	Typical Data	Key Feature(s)	Detection Strategy	Latency	Strengths	Weaknesses
GOES threshold 【13†L763-L770】	SXR flux (1–8 Å)	Flux level, monotonic rise	Requires 4-min $\wedge$ $\geq 40\%$ rise (e.g. C/M/X thresholds)	~2–5 min after onset	Simple, operational, reliable for large flares	Misses gradual rises, fails in high background, slow
RHESSI threshold 【28†L11-L19】 【28†L20-L29】	HXR counts (6–12 keV)	Count rate $>3\sigma$, detector ratio	60 s background, 3σ excess + front/total ratio	~seconds once signal emerges	Catches weaker (A–C-class) flares, uses HXR	Needs position, ignores small bursts, raw counts only
Bayesian Blocks 【31†L173-L182】	Photon time series	Likelihood of change-points	Optimal segmentation with on-line trigger mode	Near-instant (in theory)	Statistically rigorous, low false alarms if priors correct	Requires Poisson assumption, non-trivial tuning
Wavelet triggers	SXR/HXR series	Multi-scale wavelet power	Monitor short-timescale wavelet energy	Moderate (seconds, depends on FFT)	Sensitive to diverse timescales, denoising	Computationally intensive, no standard flare shape filter

Method	Typical Data	Key Feature(s)	Detection Strategy	Latency	Strengths	Weaknesses
Matched filter	SXR/HXR series	Template flare profile	Cross-correlation with flare template	Depends on template length	Theoretically optimal for known shape	Flare shapes vary; poor sensitivity to atypical flares
Derivative trigger	SXR/HXR series	Time derivative dF/dt	Trigger on dF/dt exceed threshold	Very low (seconds)	Very fast response to sharp rises	Noisy; can false-trigger on spikes
Entropy / statistical	SXR/HXR series	Entropy, variance, skewness	Detect statistical anomalies	Moderate (seconds–minutes)	Captures unpredictability	Poorly studied for flares, weakly motivated by physics

Machine-Learning Based Nowcasting

Recent work has explored classical ML models (RF, XGBoost, LightGBM, SVM, Logistic Regression) to classify or regress flare activity. These models typically ingest engineered features from prior flux or magnetic data.

- **Random Forest (RF):** RFs have been used for both forecasting and nowcasting. For example, Reep & Barnes (2021) applied an RF regressor to GOES XRS flux features to predict the *remaining duration* of an ongoing flare, achieving a median error ≈ 2 min after the SXR peak [83†L512-L522]. In classification tasks, RFs are popular due to robustness and feature importance output. Reported

performance varies (TSS ≈ 0.6 – 0.8 for 24h forecasts) depending on data

【83†L512-L522】 【92†L72-L80】 . Operationally, RFs are fast at inference and can run on modest CPU hardware. They are relatively interpretable via tree-based feature importance (and SHAP values). However, RFs can overfit if too deep, and their performance depends critically on quality of input features (they cannot automatically learn raw time-series patterns).

- **XGBoost/LightGBM:** Gradient-boosted trees (XGB, LGB) are similar to RF but often achieve higher accuracy at the cost of more tuning. Space-weather applications have reported strong performance with these (e.g. Kuleli et al., 2022 using magnetic features). They can handle imbalanced classes and give feature importances. In practice, XGBoost models have outperformed simpler tree ensembles on flare prediction benchmarks, at the expense of slower training. Real-time inference is still fast. There is little literature on XGBoost specifically for *nowcasting*, but its use in forecasting suggests it could incorporate combined SXR/HXR features. Interpretability is moderate (trees, SHAP).
- **Support Vector Machines (SVM):** SVMs with kernels have appeared in several forecasting studies (e.g. Bobra & Couvidat 2015 on HMI data). SVMs can achieve comparable skill to trees (TSS ~ 0.7) 【83†L512-L522】 . They handle high-dimensional features, but kernel evaluation can be computationally heavy for large datasets. SVMs are less interpretable (kernel weights opaque). Operational feasibility is fair for small feature sets; scaling to big data is challenging. We found no SVM-based *nowcasting* specifically; their strength in small-sample regimes may help when training on limited events.
- **Logistic Regression:** A simple linear classifier often serves as a baseline. It requires minimal computation and offers full transparency of weights. However, solar flare patterns are highly non-linear, so pure logistic models perform poorly on their own. They have been largely supplanted by more powerful classifiers. One could use logistic regression on pre-computed features (e.g. Fourier or principal components of flux), but literature on pure logistic flare models is scarce.

Features Used: Across these ML studies, typical features include SXR flux and derivative metrics (rise rate, acceleration) 【44†L510-L517】 , HXR counts or hardness (if available), and/or magnetic parameters (SHARP AR indices). For example, Reep & Barnes used flux derivatives at fixed times, and Panos & Kleint (2020) used IRIS Mg II flare signatures with DNN 【83†L503-L511】 . Many models also incorporate historical flare history of the active region. **Operational Feasibility:** Tree-based models (RF/

XGB) can be deployed in real time with low latency; SVMs similarly moderate. They require feature computation (which for XRS/GOES is trivial). Interpretability: RF and boosting can yield feature importance scores; logistic gives direct weights.

Performance: Reported skill (for forecasting) is modest (TSS ~0.5–0.7). For classification at flare onset, we lack standard benchmarks. Reep & Barnes demonstrate <3-min error for flare duration, showing RF can capture flare timing 【44†L510-L517】. In general, ML models achieve better performance than trivial climatology or persistence, but still suffer false alarms without strong physical constraints.

Deep Learning Based Nowcasting

Deep Neural Networks can ingest raw data or images and learn complex features. Key architectures:

- **CNNs:** Convolutional networks can process spatial data (e.g. solar images) or treat time-series as 1D images. Studies (Yang et al. 2021) have applied CNNs to SHARP magnetograms to predict flares, achieving high TSS (~0.8) 【92†L72-L80】. A CNN can also be applied to multi-channel time series (e.g. stacking SXR and HXR as “channels”). CNNs are fast at inference and scale well, but may require GPUs for training. They extract local temporal patterns (through 1D conv) or spatial patterns (2D conv on images). **Input representation:** Either raw flux sequences or derived spectrograms. **Window size:** Depending on convolution depth, typically minutes to hours. **Cost/Latency:** Moderate GPU training; inference can be real-time (<1 s per window).
- **LSTM/GRU (Recurrent Nets):** These process sequential data and maintain memory. They are well-suited to variable-length input. For example, an LSTM trained on 24 hr of AR features achieved TSS~0.90 【92†L72-L80】. In practice, for nowcasting one might use shorter windows (e.g. last 5–15 min). LSTMs capture temporal dynamics well, but have higher computation per step and can be slower at inference (especially long sequences). They also need careful tuning to avoid vanishing/exploding gradients. **Real-time suitability:** Good for streaming data; one step of LSTM per second is feasible on modern hardware. **Computational cost:** Higher than CNN for similar sequence length. **Detection latency:** LSTM requires the full input window, so latency = window length + model runtime.
- **TCN (Temporal Convolutional Networks):** TCNs use dilated 1D convolutions to achieve long memory. They are fully feed-forward (no recurrent states) and parallelizable. In flare contexts, e.g. the EoFNets study proposes TCNs for

predicting AR flares [96†L1-L4] . TCNs can be deep to capture hours of history with low-latency (parallel conv). They require fixed-length windows but can be made causal. **Suitability:** Very good for real-time (convolution is fast and GPU-friendly). **Latency:** Essentially convolution time (milliseconds). **Cost:** Training may be heavy if very long dilations.

- **Transformers:** Attention-based models can handle sequences and capture global context. A Transformer could attend over entire SXR/HXR time windows to spot onset patterns. However, standard Transformers are heavy (quadratic in sequence length) and data-hungry. Hybrid models (e.g. time-series Transformer with causal mask) exist, but we found no solar-flare-specific use yet. **Latency:** Generally higher (self-attention). **Pros:** Can learn long-range dependencies; well-suited to multi-modal input by multi-heads. **Cons:** Computationally costly, may be overkill for real-time alerts.
- **Hybrid Architectures:** For instance, a CNN front-end followed by an LSTM or Transformer. Or separate CNN streams for SXR and HXR followed by fusion layers. Such hybrids can capture both local (conv) and long-term (recurrent/attention) features. They are flexible but even more complex.

Deep Learning – Summary: Table 2 compares deep models. The literature shows high detection skill when ample data is available [92†L72-L80] , but many DL models have been used for *forecasting* (24 hr) rather than flare-onset detection. For true nowcasting, one needs very short windows; thus simpler nets with lower latency (1D-CNN or TCN) may suffice.

Table 2: Deep-learning models for flare nowcasting.

Architecture	Input	Example Study	Window (approx.)	Key Strengths	Weaknesses
CNN (1D/2D)	Time-series or images	Yang et al. 2021; Nishizuka 2018	Minutes–hours	Fast inference, local pattern detection [92†L72-L80]	Needs many training examples, spatial context not used if 1D

Architecture	Input	Example Study	Window (approx.)	Key Strengths	Weaknesses
LSTM/GRU	Time-series	Yang et al. 2021; Reep & Barnes†	Minutes–hours (e.g. 24h trained)	Captures temporal dynamics well 【92†L72-L80】	Slower inference, risk of forgetting older context
TCN	Time-series	EoFNets (2022)	Minutes–hours	Parallelizable, long memory, stable gradient	Less common, needs careful dilation design
Transformer	Time-series	(None in flares published)	Minutes–hours	Powerful global attention, multi-modal fusion potential	Very high compute, requires large data
CNN+LSTM (hybrid)	Images + Time-series	–	–	Combines spatial and temporal pattern learning	Very complex, long training time

Operational Space-Weather Systems

Current operational systems rely heavily on GOES SXR and predefined thresholds. NOAA/SWPC “flare alerts” are essentially generated when GOES XRS crosses thresholds (e.g. issuing R-scale bulletins when M/X flares occur). In practice, GOES XRS data are streamed in real-time, and a flare is flagged within minutes of onset (GOES needs ~4 min of rise 【13†L763-L770】). After identification, SWPC posts alerts (e.g. “M6.4 flare at 0843 UTC”). These alerts typically come out within ~5–10 min of flare

start. **Bottlenecks:** The reliance on SXR means alerts are inherently delayed by the slower rise of hot plasma. GOES detection also suffers from “flare obscuration”: a preceding large flare elevates background, masking smaller flares 【13†L763-L770】. SWPC’s *Radio Blackout* scale requires reaching M-class (R1) or higher, so smaller flares trigger no alerts at all. There is no official *pre-peak* alert; all SWPC flare “alerts” are essentially notifications of flares already occurring or past.

NASA does not issue official flare alerts, but its research satellites (e.g. Solar Dynamics Observatory) collect data that inform forecasts. Space-weather forecasts (e.g. NASA’s Community Coordinated Modeling Center products) use photospheric magnetic maps (like HMI SHARP indices) to predict daily flare probabilities 【83†L500-L509】. These forecasts are statistical and not used for immediate alerts. The Solar Orbiter/STIX instrument provides hard X-rays in near-real-time, but data latency (downlink delays of hours) and one-sided vantage (far from Earth) mean STIX is not yet used in operational alerts.

ESA’s Space Situational Awareness (SSA) program includes a flare alert system using PROBA2’s LYRA instrument (Lyman- α and EUV sensors). The LYRA Flare Finder (LYRAFF) algorithm flags flares based on a 1% increase in the 121 nm channel and other criteria 【69†L67-L75】. LYRAFF was designed to improve upon GOES event lists by separating temporally close events: e.g. it resolved ~800 double flares (2.6% of events) that GOES conflated 【69†L67-L75】. Alerts from LYRA/ESA are experimental (often emailed to subscribers) and usually coincide with GOES triggers. The advantage is slightly faster response to rapid EUV increases; the drawback is that LYRA’s low photon count can be noisy.

JAXA and other agencies currently have no autonomous real-time flare alert systems beyond standard GOES usage. In summary:

- **Detection:** Primarily via GOES XRS thresholds 【13†L763-L770】; some use LYRA or RHESSI in research.
- **Alert timing:** Typically tens of seconds to minutes after flare *start* (due to rise-time requirements). E.g. Telikicherla et al. report ML detection ~18 min before NOAA’s formal alert for big flares 【59†L90-L92】, implying NOAA’s M5 threshold alert would be ~18 min later.
- **Bottlenecks:** GOES’s slow rise, background obfuscation 【13†L763-L770】, and lack of spectral info. Also, human-confirmation steps in SWPC bulletins add delay. There is no current use of HXR data (e.g. from Fermi or SWIS) to trigger flares in real time.

Feature Engineering Review

We catalog features from the literature that can be extracted from soft and hard X-ray data. For each feature we note its meaning, computational cost, and evidence of usefulness.

Soft X-ray (SXR) Features

Typical SXR inputs are the GOES XRS 0.5–4 Å (short, A channel) and 1–8 Å (long, B channel) fluxes, or equivalent SoLEXS measurements. From these, features include:

- **Flux** (absolute irradiance): Directly measures thermal emission. Larger flux indicates stronger flare but has high background. Used ubiquitously (baselining). Low cost. However, alone it cannot predict onset — it only jumps up *after* heating.
- **Flux Derivative (dF/dt)**: Time rate of change of flux. Due to the Neupert effect, $d(SXR)/dt$ often correlates with HXR (nonthermal electrons). A sharp dF/dt signals the impulsive phase onset [37†L68-L72]. Computationally trivial. Helps flag rapidly-rising flares early. Telikicherla et al. exploited slopes of GOES flux to define HOPE signatures [59†L85-L92].
- **Flux Acceleration (d^2F/dt^2)**: The second derivative. Could indicate a change in growth trend (convexity). Rarely used explicitly; minimal extra cost once dF/dt is computed. Might catch “soft knee” structures but can be noisy.
- **Rise Rate / Onset Slope**: Often quantified as $(F_{\text{peak}} - F_{\text{start}})/\text{rise_time}$. Captures how quickly the flare intensifies. Used as a proxy for impulsiveness. Computation requires identifying start and peak (which is what one wants to do).
- **Curvature / Shape Metrics**: Fit the rising phase to a polynomial or exponential: curvature or fitting residuals can be features. Moderately costly (fits in sliding window). No standard evidence found, but it encodes whether the rise is smooth vs jagged.
- **Running Variance / Standard Deviation**: Over a recent time window (e.g. 1 min). Measures flux variability. Helps detect microbursts or flickering before a flare. Low cost ($O(\text{window})$). Could catch pre-flare turbulence; some studies of pre-flare dynamics suggest variability increases before major flares. No specific citation in literature, but as a statistical monitor it may reduce false alarms by characterizing quiescent noise.

- **Running Entropy (Shannon or Sample Entropy):** Quantifies signal unpredictability over a window. Pre-flare SXR tends to be low-entropy (steady background) but may show rising entropy as energy is injected. Computing sample entropy (MSE) is moderate cost. This has been used for magnetospheric indices 【79†L8-L12】 and might apply, but has not been widely used in flare literature.
- **Wavelet Energy:** Compute the wavelet transform (e.g. Morlet) of the SXR time series and sum the squared coefficients over given scales (especially short timescales $\sim 1\text{--}10$ s). Wavelet energy peaks would indicate bursts (e.g. microflares). This is moderately expensive (FFT-like). It has been used to identify QPP and burstiness in flares 【76†L1-L4】. As a nowcasting feature, sudden increases in wavelet energy at short scales could signal an approaching flare; not reported in detection studies but promising for multi-scale analysis.
- **Power Spectral Density (PSD) Features:** The slope or breaks of the PSD (flux vs frequency) can characterize the noise vs signal structure. For example, solar flare HXR often shows power-law PSD. Computing PSD (via FFT) over windows (say 30 s) is moderate cost. If the low-frequency power rises (steady heating) or high-frequency power spikes (impulsive bursts), this could be a trigger. This is seldom used in real-time flare work.
- **Spectral Slope:** If soft X-ray spectra (e.g. from SoLEXS's spectrometer) are available, the fitted thermal spectral slope or plasma temperature is a feature. Telikicherla et al. and others explicitly compute GOES-based temperature and emission measure 【74†L98-L107】. A rising temperature or emission measure is diagnostic of flare onset. Neubert & Battaglia (2023) and others showed that in many flares the onset phase already has $T > 10$ MK 【37†L68-L72】. Thus, an early jump in fitted temperature is a strong flare signature. Computation requires spectral fitting (moderate cost), but yields physics-based features.
- **Flux Ratios:** Ratio of short/long channel (hardness ratio in SXR) is proportional to temperature in thermal context. A rising ratio means increasing temperature. Similarly, dividing HXR and SXR fluxes (Hardness ratio) would measure nonthermal fraction (Neupert effect). Easy to compute. Used implicitly: e.g. GOES temperature is derived from SXR ratio 【37†L68-L72】.
- **Temporal Gradients:** Differences $F(t) - F(t - \Delta t)$ for small Δt , or multi-step differences. Similar to dF/dt , but discrete. Can form a vector of such gradients as features.

- **Entropy Metrics:** Beyond running Shannon entropy, one can compute “flare-like entropy” measures (e.g. sample entropy of the flux distribution) or fractal dimension/Hurst exponent of the time series. These quantify complexity or memory. In general, pre-flare SXR may show fractal-like fluctuations; values of these metrics might shift as a flare begins. These are more experimental features; few solar flare studies have directly used fractal/Hurst analysis for detection.

In summary: Features like flux, its derivatives, and spectral-fit temperature/emission measure have clear physical meaning (energy release and plasma heating) and are low-cost. They are backed by the *Neupert effect* ($\text{HXR} \approx d\text{SXR}/dt$) and hot-onset findings [37†L68-L72]. More exotic features (wavelet power, entropy) capture signal irregularity and multi-scale structure; these could help detect subtle precursors but require more computation and have less established evidence. The literature on nowcasting specifically points out the value of *onset heating* signatures (e.g. first SXR flux increase and temperature rise [37†L68-L72] [74†L100-L109]).

Hard X-ray (HXR) Features

Hard X-ray instruments (HEL1OS on Aditya-L1, RHESSI, Fermi/GBM, etc.) measure nonthermal bremsstrahlung. Key HXR features include:

- **Count Rate (Flux):** The raw HXR photon count (in specific energy bands). A sudden rise in HXR counts is the classic signature of flare onset (nonthermal electron injection). High cost? (just reading telemetry). This is the primary flare indicator.
- **Burst Duration:** How long the HXR count remains elevated above background (e.g. t_{90}). Short bursts (<1 min) often indicate impulsive flares. Once one detects a candidate burst, duration is easy to measure. Short durations often correlate with harder spectra and strong flares.
- **Energy Channel Ratios (Hardness Ratios):** Ratio of counts in a high-energy band to a low-energy band (e.g. 30–50 keV / 12–25 keV). This is a proxy for the electron energy distribution: higher ratio = harder spectrum = more energetic electrons. Hardness tends to be greater in larger flares. This is quick to compute from multi-channel data. Literature (Nagasawa et al. 2022) shows hardness varies during flares and has predictive value for subsequent evolution; as a feature it can flag nonthermal onset.
- **Spectral Hardness Index:** Fitting a power-law to the HXR spectrum yields a spectral index. Lower index (flatter) indicates stronger acceleration. Determined

via forward-fitting (expensive) or hardness ratio (fast). Strong flares often show a clear nonthermal power-law tail, so detection of a nonthermal component (or hard index) is a binary indicator of significant flare onset.

- **Nonthermal Signatures:** For example, the presence of line features (e.g. 2.223 MeV neutron capture) or higher-energy (>100 keV) emission distinguishes large flares. At nowcast onset, detecting >50 keV counts is a strong indicator of a significant flare. Such features (nonthermal tail, gamma lines) are practically only present in $>M$ -class flares, so their detection flags major events.
- **Burst Intermittency / Spikiness:** Flares often produce rapid sub-second spikes or Quasi-Periodic Pulsations (QPP). A high variance or kurtosis of the HXR lightcurve at short timescales can signal an impulsive flare. Detecting such intermittency (e.g. via running variance on sub-second bins) is computationally light and may reduce false alarms from slow variations. The Hurst exponent or fractal dimension of the HXR time series might also quantify burstiness (not used in published flare detection, but applied in other signal contexts).
- **High-Frequency Variability:** Similar to wavelet energy: the existence of significant power at frequencies >0.5 – 1 Hz in HXR suggests rapid energy release [76†L1-L4]. Calculating a short-time Fourier or wavelet transform over the HXR series can quantify this. This can detect very impulsive onsets missed by slower detectors. The cost is moderate (FFT). Research on RHESSI data shows flares have timescales down to ~ 0.1 s; capturing that in real time could give a <1 s warning if onboard processing were possible.

Strong Indicators of Onset: In practice, the very first HXR increase (even at low energies) is the earliest sign of a flare's impulsive onset. Indeed, the “hot X-ray onsets” paper [37†L68-L72] reports that HXR (STIX) detected heating even *before* GOES saw anything. Thus, any hard X-ray count above background, especially with a sudden rise or hard spectrum, is the strongest flag. Features like hardness ratio or spectral index refine the classification (e.g. distinguishing C-class vs X-class). Conversely, features like long duration or slow drift are *less* indicative of a new flare (could be decaying tail of a previous event).

Computational Cost: Extracting HXR count-rate and simple ratios is trivial. Real-time spectral fitting (to get index) is heavier but can be approximated with pre-computed lookup from ratios. Wavelet/FFT features cost more. Overall, HXR feature extraction is mostly low-latency relative to flare timescales.

Combined Soft+Hard X-ray Features (Neupert Effect)

The Neupert effect posits that the time integral of HXR (nonthermal) flux matches the SXR (thermal) flux profile. In practice, one often sees that peaks in HXR emission align with steep rises in SXR. Exploiting this, combined features include:

- **HXR vs $d(\text{SXR})/dt$ Correlation:** Compute the cross-correlation (or time-lag) between HXR lightcurve and d/dt of SXR. A high correlation (especially near zero lag) indicates Neupert behavior. One could use the peak correlation as a feature. Telikicherla's HOPE method essentially looks at when SXR rises *before* HXR appears [74†L91-L99] ; one could invert this to require that HXR precedes or coincides with SXR rise for real flares.
- **Lag Features:** Simple features like “time lag where HXR peaks relative to SXR onset”. In many flares HXR leads SXR by seconds to minutes (Neupert lag). If one detects HXR first, that implies a true flare onset.
- **Cross-Correlation Coefficients:** Over a sliding window, compute Pearson cross-correlation between SXR and HXR series. A sudden drop in correlation could signal decoupling (and a new event starting). This is medium cost ($O(\text{window})$).
- **Dynamic Time Warping (DTW) Distance:** Compute DTW between recent SXR and HXR segments. A small distance suggests the classical Neupert pattern; large distance might indicate multiple overlapping events. This is heavy ($O(N^2)$) so likely not used real-time.
- **HXR/ SXR Flux Ratios (Hardness):** Using HXR (e.g. $>25\text{ keV}$) and SXR ($1\text{--}8\text{ \AA}$) flux, the ratio or difference can be a feature. When HXR is relatively large, the flare is very impulsive (and early).
- **Plasma Energy Partition Features:** If a spectral fit yields thermal vs nonthermal components, the ratio of nonthermal energy (from HXR) to thermal emission (from SXR) is a high-level feature. Large ratios indicate strong impulsivity. Computing this requires spectroscopy and is expensive, but has clear physics meaning.

Neupert-based features in literature: Hudson (2025) explicitly proposed using GOES-derived temperature/EM and HXR together for alerting [59†L85-L92] . The HOPE concept [37†L68-L72] [74†L91-L99] is essentially combined-feature oriented (elevated SXR with no HXR yet). We therefore list Neupert-related features as highly promising: they encode direct physical causality (accelerated electrons heating plasma)

and often peak earlier than SXR alone. The challenge is aligning the two data streams and handling lags, but even a simple correlation or ratio can boost early detection.

Time–Frequency Features

Time-frequency transforms capture signal characteristics across both time and scale. Features include:

- **Fourier (FFT) Coefficients:** Compute FFT of recent data window and use magnitude at select frequencies (e.g. low vs high). If bursty high-frequency power rises, that could indicate flare activity. FFT is very fast ($O(N \log N)$). It loses time localization, so typically used as averaged features (e.g. 0–1 Hz power, 1–10 Hz power, etc).
- **Short-Time Fourier Transform (STFT):** Sliding FFTs in overlapping windows yield a spectrogram. One could track how spectral power evolves. For example, an increase in high-frequency band power during the current window might signal an onset. STFT is heavier ($O(N^2)$ if not optimized), but manageable for moderate window sizes.
- **Wavelet/CWT:** Continuous wavelet transform (e.g. Morlet) gives a time-frequency map. Useful to isolate transient bursts. For detection, one could sum wavelet power over chosen scales to get a few scalars (similar to wavelet energy discussed above). Wavelet transforms of X-ray time series have identified quasi-periodic pulsations in flares [76†L1-L4]. For nowcasting, increases in wavelet power at short scales could serve as triggers. Cost is moderate (like convolution at many scales).
- **Hilbert–Huang Transform (EMD):** Empirical mode decomposition extracts intrinsic mode functions (IMFs). One could monitor the highest-frequency IMF amplitude as a feature. EMD is computationally heavy and not widely used operationally. We found no solar-flare detections using EMD in literature.

Successful use: To date, time-frequency analysis has mostly been used in post-event analysis (e.g. QPP detection [76†L1-L4]), not as a primary detector. However, its ability to highlight very short timescale signals suggests potential: if a network measured in real time a sudden high-frequency signature (even if small), that could trigger a flare alert earlier than raw flux. There are no well-cited flare-detection papers explicitly using wavelet or STFT alarms, but the methodology is mature in other transient detection fields (gamma-ray bursts, etc).

Statistical Features

General statistics over a moving window can characterize pre-flare conditions:

- **Mean Flux:** Recent average flux. By itself not useful (background level dominates). But used as baseline subtraction (e.g. flux above mean).
- **Variance/Standard Deviation:** Measures variability. An increase in variance (over, say, the last 30 s) indicates spikiness. Low cost ($O(\text{window})$). Could help filter out “quiet” periods.
- **Skewness/Kurtosis:** High skew or kurtosis of flux distribution over a window indicates presence of outliers (spikes). Calculation is heavier but doable ($O(N)$ per window). Useful for flagging flares, since pre-flare flux has roughly Gaussian fluctuations, whereas flares produce heavy-tailed deviations.
- **Entropy Measures:** (Approximate entropy, sample entropy). Low entropy suggests predictable (quasi-periodic) flux, high entropy suggests chaotic fluctuations. If entropy suddenly rises, that may prelude a flare. Calculating sample entropy is $O(N^2)$ in naive implementation, but with fast libraries it can be done. Not common in flare literature; more used for heartbeat or EEG signals. Possibly useful to reduce false alarms: a genuine flare has lower-entropy (structured) rise than random spikes.
- **Fractal Dimension / Hurst Exponent:** Quantifies “memory” of the time series. A Hurst exponent $\neq 0.5$ suggests persistent ($H > 0.5$) or anti-persistent behavior. Flares might change the Hurst exponent of the flux (briefly). Estimation of Hurst can be done by rescaled range (R/S) analysis in $O(N \log N)$. There is some work on fractal behavior of flare time series, but not specifically for detection.

Usefulness for Onset Detection: These statistical descriptors are mostly indirect. In practice, a spike in skewness or drop in entropy during the impulsive rise could supplement other triggers, but by itself they lack physical causality. They might help to reject noise (e.g. random fluctuations have different skew/entropy than true flares). We found no specific citations using, say, skewness to alarm flares. They remain potential features to augment, rather than primary triggers.

Novel/Under-explored Feature Opportunities

From the above, we identify gaps where features have been sparsely studied:

- **Soft–Hard X-ray Coupling Features:** Beyond simple Neupert correlation, one could derive features from joint time–frequency analysis of SXR vs HXR (e.g. 2D

wavelets or coherence). For example, compute the cross-wavelet transform between SXR and HXR; power at certain scales (tens of seconds) may spike before a flare. Such multi-modal wavelet features are rarely used in the literature.

- **Multi-Scale Onset Signatures:** Most studies use fixed short windows. A promising idea is to combine very short (sub-second) and longer (tens-of-seconds) scale features. For instance, use a scattering transform or multi-scale convolution. The presence of very rapid microbursts (if any) at low amplitude just before a larger event could be a precursor.
- **Energy Transfer Signatures:** Instead of raw counts, compute the instantaneous energy flux (e.g. integrating X-ray spectra). Track the ratio of thermal to nonthermal power input. Features like sudden increase in thermal energy (from SXR) without corresponding nonthermal signature might indicate “slow heating” flare precursors (hot onset). Few works explicitly compute real-time energy budgets.
- **Pre-flare Microbursts:** Analyze very small HXR spikes (below GOES M-class scale) as precursors. Some flares exhibit “microflares” in HXR minutes before the main event. A feature could be the occurrence count or integrated energy of sub-threshold HXR bursts in a preceding window.
- **Spectral Evolution Features:** Instead of static hardness, model how the HXR spectral index changes in time (e.g. $\partial \text{index} / \partial t$). A rapid softening (soft-hard-soft pattern) is characteristic of flares; capturing the rate of change of spectral index could pre-indicate a flare.
- **Ensemble/Uncertainty Indicators:** Bayesian neural networks or ensemble variance could indicate the model’s confidence in a flare onset. A novel feature is the predicted uncertainty itself: for example, if an ensemble of predictors suddenly disagrees (high variance), that could flag an anomalous rise.
- **Graph-based Plasma Features:** If multiple detectors (like different HXR channels or multiple SXR channels) are viewed as nodes, one could construct graph features (e.g. mutual information or graph Laplacian energy). This is a speculative idea with little prior use.

In summary, many traditional features have been mined (flux, derivatives, temperature, hardness), but **truly multi-modal and multi-scale composite features** (Neupert coherence, cross-wavelets, microburst statistics) remain underexplored. These may hold potential for earlier or more reliable flare detection.

Supplementary Data (beyond Aditya-L1 SXR/HXR)

The hackathon allows extra data. We evaluate three: SWIS (proton), SDO/HMI magnetograms, and SDO/AIA EUV images.

SWIS (Solar Wind Ion Spectrometer) Proton Data

Physical justification: SWIS measures solar wind protons/alphas at L1. While solar flares often accelerate protons (SEPs), these arrive at Earth with delays of tens of minutes to hours. Thus, SWIS proton flux does *not* increase at flare onset; instead it monitors ambient solar wind. However, real-time increases in >100 keV proton flux (measured by GOES or DSCOVR) are sometimes used as post facto flare / SEP markers. For *nowcasting*, protons are too slow.

Latency and Implementation: SWIS data is essentially real-time (like ACE/DSCOVR), but for proton events it suffers propagation delay. It would at best detect a flare-driven proton spike after tens of minutes. Therefore, SWIS protons are *not useful for immediate flare detection*. One could include them to help classify flare aftermath (impact) but not onset. No prior literature uses solar-wind proton flux as a predictor for flare onset.

Conclusion: SWIS proton data has *low scientific value* for flare nowcasting (ordered rank: low). Novelty is medium (using L1 protons for any flare purpose is uncommon), but ease of use is high (real-time stream). Overall, it ranks very low for our objective.

SDO/HMI Magnetograms (SHARP parameters)

Physical justification: Photospheric magnetic field structures are the *root cause* of flares. Many studies use HMI-based parameters (flux, gradients, current helicity, shear angle, free energy proxies) to predict flare likelihood (24 hr). These Solar Dynamics Observatory (SDO) data are readily available, and SHARP (Space-weather HMI Active Region Patches) provides pre-computed indices.

Added Value: For nowcasting (immediate detection), the magnetic field is fairly static on minute timescales, so SHARP features do not change at flare onset. Their value would be in providing *context*: e.g. focusing on ARs that are “flare-productive” (high free energy) means the prior probability is higher. In ML, SHARP inputs can help avoid false alarms from magnetically weak regions.

Practicality: SHARP parameters are heavy: HMI vector data is ~ 1 MB per 12 min map, and computation (for new ARs) is non-trivial. For a hackathon with time constraints, using published SHARP tables is easier but those have several-hour latency (mostly

they are available within an hour of observation). However, since Aditya-L1 data is near-real-time, one could still incorporate the latest SHARP snapshot for each AR as an auxiliary input.

Literature: SHARP features are proven predictors in forecast ML (Bobra & Couvidat 2015; Nishizuka et al. 2018), but almost never used for nowcast. They are largely ignored by flare-detection literature because they do not provide temporal signals.

Ranking: SHARP parameters have *high scientific value* (strongly correlated with flare energy) but *low novelty* (widely used in flare forecasting). Implementation is moderate difficulty (data access, feature extraction). For the hackathon, they could be included as static features per AR, but as indicated they will likely not improve the *timeliness* of onset detection.

SDO/AIA EUV Images

Physical justification: The Atmospheric Imaging Assembly (AIA) on SDO provides high-resolution full-disk EUV images at multiple passbands (e.g. 131 Å, 171 Å) at 12 s cadence. Pre-flare EUV brightening or coronal dimming can precede flares. e.g. filament activation in 304 Å or coronal loop heating in 131 Å may show up seconds to minutes before X-rays rise. In principle, automated image analysis (CNN) could detect such patterns.

Predictive Value: Studies (e.g. Nishizuka 2018; Huang 2018) have shown AIA images add predictive power for *forecasting*. For nowcasting, very few have tried it. One could imagine a CNN processing a small cutout around an AR in a hot channel (131 or 94 Å); recent brightening there might signal an imminent flare. The catch is that EUV heating to ~10 MK often *coincides* with X-ray rise, so time lead may be small. However, subtle morphological changes (ribbon formation, coronal waves) might give extra seconds.

Computational Burden: AIA images are huge (4096²) and multi-channel (7 EUV filters). Processing even downsampled images in real time is extremely heavy (requires GPUs and fast I/O). For hackathon use, one could restrict to one channel (e.g. 131 Å, sensitive to ~10 MK plasma) and bin to low resolution (e.g. 128×128) around the AR. Real-time use is challenging.

Ranking: EUV images have *very high scientific value* (rich multi-thermal structure), moderate novelty (some ML studies exist), but *very low ease* (massive data). If computational resources allow, a lightweight CNN on EUV could be implemented (e.g. few layers on downsampled frames). However, in practice, using even one high-cadence EUV channel is daunting.

Supplementary Datasets Ranking

Data Source	Scientific Value	Novelty (for nowcasting)	Ease of Use	Comments
SDO/AIA EUV	High	Medium	Difficult	Rich pre-flare info; CNN needed; heavy data
SDO/HMI SHARP	High	Low	Medium	Proven flare predictors; static context only
GOES/XRS (added)	High	Low	Easy	Complement SoLEXS; NOAA standard
RHESSI/GBM HXR	Medium	Medium	Easy	Redundant with HEL1OS; use for validation
Fermi GBM	Medium	Medium	Easy	Real-time HXR similar to HEL1OS; archival use
SWIS Protons	Low	High (novel)	Easy	SEP lag; not useful for onset
PROBA2/LYRA	Low	Low	Medium	Already produces flare list (LYRAFF) 【69†L67-L75】

Gaps and Limitations in Existing Work

Reviewing the above literature and practices reveals several shortcomings:

- **GOES-Based Methods:** Assume a single flare and monotonic rise, so they miss overlapping/multiple flares [13†L763-L770]. They also ignore any signal below threshold (e.g. sub-C flares). No use is made of HXR or EUV precursor data. False alarms can occur from gradual background changes (cosmic rays, instrumental drift). Early detection is limited by the ≥ 4 min rise rule [13†L763-L770].
- **RHESSI Algorithm:** Requires $>3\sigma$ counts and a viable imaging solution [28†L20-L29]. Assumes flares produce a localized footpoint (for imaging). Misses flares with weak/non-existent imaging or where front/total ratio condition fails (e.g. faint thermal flares). It also fails if decimation or attenuator state changes. It ignores any spectral information beyond count rate.
- **Bayesian Blocks / Change-Point Methods:** These assume data as Poisson counts and aim to fit constant-rate blocks. Solar X-ray backgrounds violate simple models (varying A/C channel baseline). They may detect any change (including gradual trends) as flare, causing false positives. Also, tuning the “false positive probability” parameter is nontrivial (too sensitive $\rightarrow$ floods, too conservative $\rightarrow$ misses). Gaps in data or varying exposure can break the assumption.
- **Wavelet/Matched Filters:** Literature has mostly used wavelets for analysis, not for real-time triggers. Proposed filters often assume a generic flare shape, which is oversimplified. No systematic evaluation exists of detection latency vs false alarms for wavelet triggers. Moreover, these methods have not incorporated multi-channel data (SXR+HXR) jointly.
- **Machine Learning (RF, SVM, etc):** Many studies focus on *forecasting* (24 h ahead) using SHARP/magnetogram features. Assumptions include that the magnetic configuration contains sufficient predictive information. These neglect immediate flux measurements (SXR/HXR), thus failing to *detect* flares as they begin. ML models often ignore the Neupert effect by treating SXR and HXR features independently, missing their temporal coupling. They typically assume a static feature set; dynamic temporal features (like dF/dt) have been used only as simple time-series inputs. Many ML studies also suffer from data biases (active-region selection, unbalanced classes) leading to optimistic scores. Operational

ML often requires retraining to remain calibrated to new solar cycles, a gap seldom addressed.

- **Deep Learning:** Existing DL work mostly addresses flare forecasting with long input windows (hours). These models assume a long pre-flare history is available; they also require large labeled datasets (flare vs no-flare) which bias them to stronger flares. They often ignore multi-modal fusion beyond single sources (e.g. only HMI or only SXR). Few DL models consider real-time constraints; many treat the problem offline. They also rarely focus on *latency minimization*; e.g. an LSTM trained on 24 hr data is useless for real-time 5 min nowcasting. Finally, most DL models provide little interpretability – attention maps or Grad-CAM are seldom analyzed in flare context.
- **Operational Systems:** The NOAA/ESA systems assume no precursor before SXR threshold crossing, so they fail to exploit any early heating signals. They also generate no early warnings for sub-M flares (important for aviation HF disruptions). A clear gap is the lack of any HXR-based alert; current alerts are purely SXR-based. False alarms in operational lists often come from false positives in detection algorithms (e.g. LYRAFF sometimes splits single flares incorrectly [69†L67-L75]). Bottlenecks include data latency (though GOES is <1 min, HMI is slower) and human-in-the-loop in issuing bulletins.

Table 3: Research Gaps in Flare Nowcasting.

Aspect	Common Approaches	Unmet Issues / Ignored Aspects
Assumptions	Monotonic rise, one flare	Fail for overlapping or complex flares; ignore faint precursors; assume Gaussian noise (often false)
Datasets Used	GOES SXR (NOAA), SHARP	Rarely use HXR (RHESSI/STIX/Fermi) or EUV in detection; new Aditya data not yet exploited

Aspect	Common Approaches	Unmet Issues / Ignored Aspects
Features Ignored	Static AR features, flux	Neupert coupling, multi-scale fluctuations, polarization (magnetism) during flare
Models/Methodology	Single-threshold, simple ML	Lack real-time learning; forecasting models not adapted for instant detection; few hybrid multi-modal models
False Alarms	Hard-coded rules threshold	Instrument noise and cosmic rays can mimic flares; statistical models often flag any deviation as flare
Detection Latency	$\geq$ minutes (due to smoothing)	Early phase (pre-impulsive heating) is not captured; no upstream alert ahead of GOES threshold
Generalization	Static classifiers	Often trained on past cycle; performance degrades on new cycles; few methods incorporate uncertainty estimates

Novel Aditya-L1 Nowcasting Ideas

Based on the above gaps and the unique availability of SoLEXS (soft X-rays) and HEL1OS (hard X-rays) on Aditya-L1, we propose ten innovative approaches:

1. **Neupert-Aware Dual-Stream Network:**

2. *Rationale:* Explicitly leverage Neupert effect by feeding SXR and HXR into parallel neural streams with a cross-attention layer.
3. *Architecture:* Two 1D CNN branches (one for SXR, one for HXR time series) whose outputs are fused by a transformer-like attention module that aligns features temporally. Follow with a dense classifier.
4. *Features:* Raw or normalized XRS fluxes, their short-time derivatives, HXR counts in several bands.
5. *Advantages:* Learns synergy between SXR rise and HXR bursts, automatically aligning timings. Attention weights are interpretable (which timepoints in which channel triggered the alert).
6. *Risks:* Requires careful training to not overfit on correlation patterns; attention may spread.
7. *Complexity:* Moderate-high (two CNN streams + attention).
8. *Novelty:* High – direct Neupert fusion in deep net.
9. *Hackathon Suitability:* Good if GPUs available; moderate coding complexity.

10. **Multi-Scale Wavelet Triggering:**

11. *Rationale:* Capture onset signatures across scales (from sub-second to tens of seconds) by multiresolution analysis.
12. *Architecture:* Preprocess SXR/HXR with a wavelet packet transform (e.g. db4) in real time; extract energy in selected high-frequency bands. Input these energies to an RF or LightGBM classifier.
13. *Features:* Wavelet coefficients' energy at various scales from both SXR and HXR.
14. *Advantages:* Sensitive to subtle fast events or long drifts; agnostic to exact shape.
15. *Risks:* Parameter tuning (wavelet type, scale selection) is crucial; could be noisy.
16. *Complexity:* Moderate (real-time filter bank + ML).
17. *Novelty:* Medium – wavelets are known but seldom used for triggers.
18. *Suitability:* High: wavelet filtering is fast (FFT-based) and decision trees are quick.

19. **Soft-Hard Attention Fusion (DL Hybrid):**

20. *Rationale:* Use a Transformer-like model to flexibly weight SXR vs HXR data in prediction.

- 21. *Architecture*: Concatenate SXR and HXR time series into one sequence with channel encoding. Feed through a 1D Transformer encoder. Output goes to classification head (flaring vs not).
- 22. *Features*: Raw time-series channels, optionally position-encoded.
- 23. *Advantages*: Automatically learns which times and channels are most predictive. Attention maps show whether SXR or HXR dominated the decision.
- 24. *Risks*: Transformers are data-hungry; may overfit on limited flare events.
- 25. *Complexity*: High (self-attention).
- 26. *Novelty*: High – attention for multi-modal time series fusion is cutting-edge.
- 27. *Suitability*: Challenging for hackathon timeframe, but feasible with pre-built libs (e.g. PyTorch Transformer).

28. **Physics-Informed TCN:**

- 29. *Rationale*: Embed domain knowledge (e.g. SXR-HXR causal ordering) into a Temporal CNN.
- 30. *Architecture*: A causal TCN taking as input features like [SXR flux, HXR flux, $d(\text{SXR})/dt$] that constrains convolutional kernels to respect temporal causality. Use dilations to cover up to ~60 s.
- 31. *Features*: Combined SXR and HXR fluxes and computed derivatives.
- 32. *Advantages*: TCN offers very fast inference and stable training. Being causal ensures no future leak.
- 33. *Risks*: Designing kernel sizes/dilations to capture both rapid and slow changes is tricky.
- 34. *Complexity*: Moderate (stacked conv blocks).
- 35. *Novelty*: Medium – TCNs for flare detection are rare; physics-informed flavor is novel.
- 36. *Suitability*: Good – TCN code is simpler than RNNs/transformers for short sequences.

37. **Flare State Machine (Hybrid Rules + ML):**

- 38. *Rationale*: Model flare evolution as a finite-state process (Quiet → Onset → Impulsive → Decay) and use simple rules for transitions, augmented by ML to estimate state durations.
- 39. *Architecture*: A hidden Markov model (HMM) or state machine where observations are SXR and HXR values. Emission probabilities (flux distributions) in each state can be learned via RF or Gaussian models.

- 40. *Features*: Current SXR/HXR, recent derivatives; state at previous time (initially quiet).
- 41. *Advantages*: Explicitly encodes flare physics (impulsive has high HXR, gradual SXR rise). Can force consistency over time (e.g. no jump from quiet to decay).
- 42. *Risks*: Defining the correct state transitions and training emissions may be complex. Might lag behind if state change is slow.
- 43. *Complexity*: Moderate; HMM libraries exist.
- 44. *Novelty*: High – state-machine approach is seldom used for flares.
- 45. *Suitability*: Good – conceptually clear; might require careful parameter tuning.

46. **Graph Neural Network (Temporal Graph):**

- 47. *Rationale*: Represent SXR and HXR features as nodes in a dynamic graph and learn their interactions.
- 48. *Architecture*: At each time step, create a graph node for each channel/band (e.g. SXR band, each HXR energy bin). Connect nodes (edges) weighted by physics (strong SXR-HXR edge). Use a recurrent GNN that updates node features over time.
- 49. *Features*: Channel fluxes, historical derivatives.
- 50. *Advantages*: Models multi-stream coupling explicitly; can learn how energy flows between nodes.
- 51. *Risks*: Very novel, untested for flares; data for training GNN could be limited.
- 52. *Complexity*: High (GNN + temporal).
- 53. *Novelty*: Very high.
- 54. *Suitability*: Perhaps too complex for hackathon; better as proof-of-concept research.

55. **Multi-Task CNN/LSTM (Detection + Regression):**

- 56. *Rationale*: Train a network that simultaneously (a) classifies “flare now” and (b) predicts flare parameters (peak flux, time-to-peak). Shared features will improve both tasks.
- 57. *Architecture*: A shared CNN (1D conv) followed by two heads: one sigmoid for detection (yes/no), one linear for regression (e.g. minutes to peak).
- 58. *Features*: SXR/HXR sequences, possibly aggregated statistics.
- 59. *Advantages*: Forces network to understand flare progression, not just binary event. Could help give early magnitude estimate.

- 60. *Risks*: Need to label peak times and magnitudes accurately; regression may distract classification.
- 61. *Complexity*: Moderate.
- 62. *Novelty*: Medium – multi-task learning is common in ML but not in flare nowcasting.
- 63. *Suitability*: Good, uses standard components; interpretability moderate (two outputs).

64. Uncertainty-Aware Bayesian NN:

- 65. *Rationale*: Provide not just an alert but confidence. A Bayesian (e.g. MC Dropout) or ensemble NN can estimate uncertainty in predictions.
- 66. *Architecture*: Any of the above networks with dropout at inference or as an ensemble; outputs a probability distribution for flare vs not.
- 67. *Features*: Same as chosen backbone (CNN/LSTM, etc).
- 68. *Advantages*: Crucial for operational alerts – will not fire if too uncertain (reducing false alarms). Uncertainty itself is a novel feature (spike in model uncertainty might indicate something new).
- 69. *Risks*: Calibration of uncertainty is non-trivial; high uncertainty in quiet times may not always help.
- 70. *Complexity*: Low-mod (just implement dropout or ensemble).
- 71. *Novelty*: High for this domain (few flare models report uncertainty).
- 72. *Suitability*: Good – adds little overhead but improves robustness.

73. Physics-Informed Neural Feature (PINN) Preprocessor:

- 74. *Rationale*: Use a small neural network or autoencoder to learn a physics-inspired transformation of the raw X-ray spectra (e.g. fit two-temperature model). Feed that into ML.
- 75. *Architecture*: Pre-train a network to output physical parameters (like EM, T, nonthermal index) from raw detector counts. Use those as features for a simple classifier.
- 76. *Features*: Derived plasma parameters plus raw residuals.
- 77. *Advantages*: Encodes known physics (an approximation to forward fit) without hand-coding equations. Learns to denoise instrument quirks.
- 78. *Risks*: Requires training data with spectra; may be redundant if fast spectral fitting is done anyway.
- 79. *Complexity*: Moderate (two-step model).

80. *Novelty*: Medium – PINNs are popular but rare in flare detection.
81. *Suitability*: Feasible as long as spectra for training exist (MinXSS or GOES proxies).

82. Real-Time Alarm Cascade (Expert System):

83. *Rationale*: Mimic an expert forecaster by a series of logical tests on incrementally processed data. If SXR rises slightly but HXR shows nothing, go to “watch” state; if HXR appears, issue an alarm, etc.
84. *Architecture*: A set of hierarchical rules/thresholds (a small decision tree) applied to features such as: (a) Is $d(\text{SXR})/dt > X$? (b) Has HXR exceeded Y in last Z sec? (c) Is temperature $> 10 \text{ MK}$? (d) Is model uncertainty low? If yes to all, trigger immediate alert.
85. *Features*: Direct physical indicators (flux, dF/dt , T).
86. *Advantages*: Easy to understand and tweak; can incorporate expert insight.
87. *Risks*: Hard to optimize thresholds; brittle to noise.
88. *Complexity*: Low.
89. *Novelty*: Low (heuristic rule-based), but new in combination with new instruments.
90. *Suitability*: High – can be prototyped quickly and even combined with ML (as a veto stage, for example).

Each idea aims to exploit SXR–HXR data in novel ways. For example, combining attention-based fusion (Idea 3) could outperform simple concatenation models by explicitly weighting the significance of each channel. The multi-scale (Idea 2) and multi-task (Idea 7) concepts attempt to squeeze more information out of the time series. Ideas involving uncertainty and hybrid rule-ML (6, 8, 10) enhance reliability and interpretability, which are critical for real-world alerting.

Final Recommendation

Chosen System: We recommend a **Neupert-informed Hybrid Network** as the core nowcasting system. This system fuses SoLEXS SXR and HEL1OS HXR data in a deep learning architecture that explicitly models their physical relationship.

- **Feature Pipeline:** As input, we use a short sliding window (e.g. last **60 seconds** of data at 1 Hz sampling) of: (1) SXR flux (0.1–10 keV, e.g. combined or both SoLEXS channels), (2) HXR count rates in a few energy bands (e.g. 5–15 keV, 15–50 keV), (3) the time-derivative of SXR flux. From these raw series we also

compute simple statistics: recent running mean and standard deviation, and SXR/HXR hardness ratios. This yields ~10–15 features at each time step (a multivariate time series). If spectral fits are available, we include temperature and emission measure as additional streams (low-priority).

- **Model Architecture:** We use a **Dual-Stream 1D Convolutional Network with Cross-Attention**. Concretely: two parallel 1D CNN encoders (several layers of causal convolutions with ReLU, capturing local temporal patterns) process the SXR and HXR sequences separately. Each produces a sequence of high-level features. We then apply a **cross-channel attention module**: the SXR feature sequence attends to the HXR sequence and vice versa (multi-head attention with causal masks), allowing the model to learn alignments (e.g. “if HXR spikes now, attend to SXR a few seconds later”). The attended features are concatenated and fed into a final feed-forward classifier that outputs: (a) a **flare flag** (binary: flare-onset vs quiet) and (b) a **regression** for estimated time-to-peak flux (or peak magnitude). We train the model end-to-end with a combined loss (cross-entropy + L1 regression). Dropout is used throughout for regularization and Monte-Carlo uncertainty estimation.
- **Data Fusion:** We fuse SXR and HXR both at input (as parallel channels) and mid-model (via attention). Early fusion is implicit in attending between streams, so the network learns, for instance, that an HXR increase should be weighted more if preceded by an SXR rise (and vice versa). The cross-attention weights themselves are insightful: they will highlight time lags where SXR and HXR interplay is strongest.
- **Explainability:** We will apply **SHAP values** and attention-visualization on the trained model. SHAP (tree-based adaptation to neural nets) can attribute the flare output to input features/timepoints. The attention weights provide a direct interpretability handle: they show which time steps in which channel drove the alert. We also plan to train a simpler surrogate (e.g. decision tree) on the neural-net’s decisions for human-understandable rules. In addition, we can provide the user with key feature triggers (e.g. “SXR $dF/dt > X$, HXR $> Y$ ”).
- **Real-Time Deployment:** The system operates on streaming SXR/HXR data. We maintain a rolling buffer of the last 60 s (or as configured) and run the neural net every Δt (e.g. every second). The architecture is fully causal, so no future data is used. Inference is lightweight (a few conv layers), so a modest GPU or even a high-end CPU can run it with sub-second latency. Outputs (alert flag, probabilities, predicted peak) are published immediately if above thresholds. A threshold on the output probability (tuned for desired false-alarm rate) determines

actual alarms. We implement an “arming” logic: once triggered, the system enters an “in-alert” mode to avoid duplicate flags for the same event.

- **Training Strategy:** We train on historical data labeled for flare onsets. Since Aditya-L1 is new, we use surrogate data: GOES/XRS for SXR, RHESSI/Fermi for HXR (or STIX if overlapping). We compile a dataset of past flares (with start times) and non-flare intervals. The input windows end just after the labeled flare start. We include a wide range of classes (C, M, X) and imbalanced negative examples. We perform data augmentation by adding slight noise, scaling, and by randomly shifting window alignment. The loss is weighted to penalize missing large flares most heavily. We validate by k-fold on past events (e.g. train on solar cycle 24, test on cycle 25).
- **Validation Strategy:** We will evaluate detection (Precision/Recall, TSS) and timing error (prediction vs true start) on held-out data. Cross-validation on different years tests generalization. We also simulate real-time runs on historical time series (sliding window) to compute false alarm rates. As ground truth, we use NOAA event lists and RHESSI catalogs.
- **Ablation Studies:** We will remove components to measure impact: (a) no HXR branch (to quantify X-ray coupling), (b) no $d(\text{SXR})/dt$ feature, (c) attention disabled (simple concatenation), (d) regression head removed. We expect (a) HXR removal will drop early detection skill (since hot onsets will be missed), and (b) removing derivative features will slow detection. We will report metrics for each variant.
- **Expected Performance:** Based on related studies, we anticipate high skill. Using [59†L85-L92] as a benchmark (Telikicherla’s ML caught all $X>M5$ flares with 17.9 min lead over NOAA), our model should similarly detect strong flares *well before* NOAA’s threshold alerts. We target $>90\%$ detection of $\geq M$ -class flares with minimal false alarms. Compared to existing methods (GOES threshold alone), we expect improvement in lead time (several minutes earlier) and detection of some sub-threshold events. We will measure average alert lead time relative to NOAA’s R3 alert; a few-minute gain is a key goal.

Why It Outperforms Existing Approaches: This system directly addresses known weaknesses: it uses both SXR and HXR (exploiting hot onsets [37†L68-L72]), it leverages the Neupert effect via learned attention, and it is data-driven to adapt to real flare morphology rather than fixed thresholds. The inclusion of uncertainty and regression outputs adds valuable information. In contrast, traditional detectors lack HXR

data and deep networks without physics may miss causal relations. Our architecture combines the strengths of both: physics-informed design with ML flexibility.

Architecture Diagram (text form):

- Input: [SoLEXS SXR time series (60 s window), HEL1OS HXR time series (same window)]
- ⇒ **Parallel 1D-CNN encoders** (each: CausalConv → ReLU → CausalConv → ReLU → ...) for SXR and HXR streams.
- ⇒ **Cross-Attention layer**: SXR features attend to HXR features and vice versa (temporal self-attn with causal mask).
- ⇒ **Concatenation of attended features**.
- ⇒ **Fully-Connected layers** leading to two outputs: (1) Sigmoid Flare-Flag; (2) Linear time-to-peak (or peak flux) regression.

References: We draw on solar-flare data and methods from the literature: GOES detection criteria [【13†L763-L770】](#) , RHESSI triggering [【28†L11-L29】](#) , hot-onset physics [【37†L68-L72】](#) [【74†L100-L109】](#) , ML forecasting successes [【83†L500-L509】](#) , and recent nowcasting proposals [【59†L85-L92】](#) . All algorithmic claims will be validated by reanalysis of historical flare events and cited studies (above) support our feature choices and design.
